

KESHAV KRISHNA

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GitHub | Website | LinkedIn | Google Scholar

Graduation: May 2027 | Available to start: May 2027

Software and machine learning engineer with three years of experience building production recommendation, personalization, and LLM systems for a consumer product serving 100M+ users. Experienced across the full ML lifecycle: data pipelines, model development, debugging, training, evaluation, inference, deployment, profiling, and iteration. Strong foundation in Python, C/C++, Linux, algorithms, system design, distributed data processing, and ML platform tooling.

EDUCATION

New York University, Courant Institute — MS, Computer Science

GPA 3.95/4.0; Machine Learning, Building LLM Reasoners, Honors Algorithms

Aug 2025 – May 2027

New York City

Indian Institute of Technology, Ropar — BTech, Computer Science and Engineering

GPA 8.74/10; Merit Scholarship for top academic performance

Aug 2019 – Jun 2023

Ropar, India

EXPERIENCE

Moore Capital Management — Risk Technology Intern

May 2026 – Present

Risk analytics and scalable data tooling; full-time summer, part-time fall

New York City

- Built Python decision-support tools for portfolio and risk teams in a hedge-fund environment, combining data from multiple sources into clear, actionable analytics used in daily workflows.
- Developed and benchmarked distributed data-processing tools across large-scale compute environments; used profiling and system modeling to identify scalability bottlenecks and guide infrastructure decisions.
- Optimized Value at Risk (VaR), Conditional Value at Risk (CVaR), beta, and correlation calculations with vectorized Python and Numba JIT compilation; benchmarked implementations to reduce runtime on large financial datasets.

JioSaavn — Software Engineer, Data Science

Jul 2023 – Jul 2025

Production recommendation, personalization, and language-model systems

Mumbai, India

- Built and iterated on recommendation and personalization systems using embeddings, vector retrieval, Spark, Hive, Redis, and Qdrant, driving **1M impressions, 800K clicks, and 500K streams per day** in a 100M+ user product.
- Owned the ML lifecycle for a LLaMA3-8B-Instruct conversational playlist system, from intent modeling and retrieval-driven song selection through evaluation, inference, and deployment with Docker and Kubernetes.
- Designed a reproducible evaluation across SetFit, HingRoBERTa, mURIL, and LLaMA3-8B-Instruct for multilingual moderation; built a labeled test set and selected a model achieving **85% accuracy**.
- Collaborated with product, editorial, and infrastructure teams to translate ambiguous requirements into production models, data pipelines, and services.

GE Healthcare — Software Intern, Medical Computer Vision

May 2022 – Jul 2022

Detection and de-identification pipelines for clinical imaging

Bangalore, India

- Built and evaluated YOLO-based medical-image de-identification and RGB/IR people-detection pipelines over 20K+ images.

SELECTED ML SYSTEMS PROJECTS

FlashAttention-2 — Triton GPU Kernels [GitHub]

2026

- Implemented forward and backward attention kernels with tiled parallel work partitioning, online softmax, causal masking, and activation recomputation; reduced attention memory from $O(N^2)$ to $O(N)$ and enabled 65K-token sequences.
- Built a Linux GPU profiling and benchmarking suite for Transformer models up to 2.7B parameters using NVIDIA Nsight Systems, CUDA memory snapshots, NVTX, mixed precision, and `torch.compile` baselines.

SELECTED ML SYSTEMS PROJECTS (CONTINUED)

Transformer Language Model and Training Stack from Scratch [GitHub] 2026

- Built a 17M-parameter PyTorch language model from low-level components: BPE tokenization, RoPE, RMSNorm, SwiGLU, causal attention, AdamW, gradient clipping, checkpointing, and cosine learning-rate scheduling.
- Implemented multiprocessing-based parallel tokenization, memory-mapped data loading, automated training sweeps, FLOPs and memory accounting, and systematic architecture ablations for reproducible model development and evaluation.

Pico-LLM — Training, Evaluation, and Inference Platform [GitHub] 2025

- Developed a unified platform for SFT and DPO/GRPO training, checkpoint management, batch evaluation, and inference across Transformer, LSTM, and K-gram architectures.
- Added KV-cached generation, multi-GPU batch generation, automatic architecture detection, and robust token handling to improve inference throughput and experiment reliability.

Phase-Adaptive Generation for Diffusion LMs [GitHub] Feb 2026 – May 2026

- Analyzed per-token refinement traces and trained Transformer predictors over 138K control tuples to assign dynamic per-block inference budgets.
- Integrated the predictor into the LLaDA decoding loop and matched AdaBlock’s 89.5% GSM8K accuracy with approximately **21% fewer forward evaluations**; designed experiments to validate quality-efficiency tradeoffs. [Report]

ML-Driven Hardware Resource Optimization [GitHub] Jan 2026 – May 2026

- Modeled temporal traces from multicore workloads, clustered hardware pressure states, and distilled them into compact predictors with sub-KB model state for dynamic resource-management decisions.

RESEARCH AND PUBLICATIONS

- *Advancements in Cache Management: A Review of Machine Learning Innovations* — Frontiers in AI, 2025. Single-author review of 40+ papers on ML-driven cache and resource management.
- *LLC Intra-set Write Balancing for Non-Volatile Memory Caches* — arXiv, 2024. Bachelor’s thesis research on system-level cache optimization.

TECHNICAL SKILLS

Programming and systems: Python, C/C++, JavaScript, Linux, data structures and algorithms, system design, multiprocessing, distributed systems, Git

ML development lifecycle: PyTorch, Hugging Face, vLLM, Triton, CUDA, model development, debugging, training, evaluation, inference, recommendation systems, vector search, SFT, GRPO, RLHF

Platforms and infrastructure: Kubernetes, Docker, Spark, Hive, Airflow, Redis, Qdrant, Slurm/HPC, NVIDIA Nsight Systems, Numba JIT, FastAPI, Postgres